

Proposal: on scalable partition-based spatial Gaussian Process

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Summary

Data associated with locations, known as spatial data, is used across fields such as geography, public health, and social sciences. Gaussian Process (GP) is a popular tool to model spatial data that can make predictions on the values at new locations. Traditional GP models rely on the assumptions that the mean and covariance patterns can be specified by a single mean function and a covariance function. However, these assumptions often fail in real applications. For example, there may exist a change boundary such that the data on different sides have different mean and covariance patterns. This research aims to enhance the goodness of fit on datasets with such boundaries. Following existing literature, we partition the space into smaller subspaces where a single GP model works well, and fit separate models in each subspace. However, most existing partition-based models are constructed in Bayesian settings and use heavy sampling techniques that are computationally demanding. We alternatively propose a framework of algorithms that comprises a partitioning scheme and an efficient GP fitting approach using approximation. We proposed an INLA-tree-GP model under this framework with a decision tree partition scheme and the INLA (Integrated Nested Laplace Approximation) approach. We expect the proposed method to simultaneously enjoy scalable computation and accurate predictions on large spatial datasets with complex structures.

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1 Introduction

Data associated with locations, known as spatial data, widely appears in many fields, including Geography, Social science and epidemiology. Each sample is a pair of a location and an object of interests, denoted as $(\mathbf{x}, z(\mathbf{x}))$. The location $\mathbf{x}$ is often represented numerically, using, for example, Cartesian coordinates on a plane or longitude and latitude on the globe. We denote $\mathcal{D} \ni \mathbf{x}$ to be the spatial space of interest that includes all targeted locations. Mathematically, $\mathcal{D}$ is often a subset of the Euclidean space $\mathbb{R}^d$ with dimension d or a manifold with a geometric structure (e.g., the globe). A distance function

$$\text{dist} : \mathcal{D} \times \mathcal{D} \mapsto [0, \infty]$$

is defined to quantify the proximity between any two locations. The squared Euclidean distance is a common choice. The object $z(\mathbf{x})$ is a random variable associated with $\mathbf{x}$. In many cases, only a finite number of samples $\{\mathbf{x}_i, z(\mathbf{x}_i)\}_{i=1}^n$ are available, and it is of interests to give inference to the variables at any new position $\mathbf{x}^* \in \mathcal{D}$ outside the sample. For modelling purpose, it is natural to make the following assumptions.

- (A1) The mean of an object z depends only on its associated location, modelled by a mean function f across the space of location,

$$\forall \mathbf{x} \in \mathcal{D}, \mathbb{E}[z(\mathbf{x})] \leftarrow f(\mathbf{x})$$

which can be simply zero, a constant, any given function or any parameterised function f_{θ_f} .

- (A2) The objects in close proximity are highly correlated. Thus, the covariance between $z_1(\mathbf{x}_1)$ and $z_2(\mathbf{x}_2)$ is modelled to be a function k that decreases as the distance $\text{dist}(\mathbf{x}_1, \mathbf{x}_2)$ increases. k is further isotropic, which means it depends only on the distance, i.e.,

$$\forall \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{D}, \text{cov}[z_1(\mathbf{x}_1), z_2(\mathbf{x}_2)] \leftarrow k(\mathbf{x}_1, \mathbf{x}_2) := k[\text{dist}(\mathbf{x}_1, \mathbf{x}_2)]$$

where $k := k_{\theta_k}$ are usually parameterised.

Based on these two assumptions, Gaussian Process is a popular choice to model $z(\mathbf{x})$ across $\mathcal{D}$ by further assuming a multivariate normal distribution on any finite sample. It possesses the benefits of simplicity in capturing the mean and covariance functions and convenient prediction in a conjugate

form distribution. However, the assumptions are fragile in real cases. The space $\mathcal{D}$ may be large, which means different regions may have different mean functions and covariance functions; there may be a change boundary in $\mathcal{D}$ such that the variables at different sides, even though located in close distance, differ dramatically in value and have no correlation with each other. The boundary is ubiquitous, such as a stratigraphic boundary, a cliff on a contour map or a mountain that results in great differences in rainfall. It may be of research interest to identify the boundary in addition to fitting the GP model.

We focus on spatial data with such boundaries. Furthermore, we assume that the space $\mathcal{D}$ can be partitioned into several subspaces, where the underlying data-generating mechanism shares some common features within each subspace such that the same mean function and covariance function apply, while variables in different subspaces have no correlation and come from different data-generating mechanisms.

A straightforward idea is to partition the space and fit an individual GP model in each subspace. Models with this idea have long been developed in the literature. However, most of them are constructed in the Bayesian setting, and there lacks highly efficient MCMC (Markov Chain Monte Carlo) methods designed for such a complex scenario. This puts an extra computational burden on the standard GP that has already involved a matrix inversion in $\mathcal{O}(n^3)$ time complexity. These models are computationally infeasible for a large sample size n .

The research aims to tackle this issue and develop a scalable partition-based spatial Gaussian Process model. We propose a framework for spatial partition algorithms that avoid MCMC and use an approximation approach to fit and evaluate a GP efficiently. The framework comprises a partition scheme and an efficient GP fitting approach. We will develop a specific model under this framework based on a decision tree and Integrated Nested Laplace Approximation (INLA). The model is expected to outperform the current benchmarks in computation time and model fitness.

The objectives of the proposed research are stated in the following section. The proposal will proceed to review the existing literature in partition-based GP models before elaborating our proposed framework and model that are expected to achieve the objectives.

2 Research objectives

- (O1) Design appropriate partition schemes that account for both computational feasibility and prediction accuracy.
- (O2) Use approximation methods for computational efficiency that are suitable for partition schemes given in (O1).
- (O3) Develop scalable partition-based models by considering (O1) and (O2). Implement them in Python or R packages.
- (O4) Match or surpass the state-of-the-art methods in terms of both computational cost and prediction accuracy.

3 Literature review

This section begins with the setting of spatial Gaussian Process models. Then it reviews some existing partitioning schemes and discusses their strength and limitations. Lastly, it introduces an approximation-based scalable GP approach that is applicable to our proposed methods.

3.1 Gaussian Process model

We begin with some notations for Gaussian Process. We denote the Gaussian process as

$$z \sim \mathcal{GP}_{\mathcal{D}}(f, k)$$

$\mathcal{D}$ is the location space. In the scope of this proposal, we simply assume $\mathcal{D}$ to be $\mathbb{R}^2$ or a rectangle in $\mathbb{R}^2$. A distance function $\text{dist} : \mathcal{D} \times \mathcal{D} \mapsto [0, \infty]$ is defined. By assumption (A2), $k(\mathbf{x}_1, \mathbf{x}_2)$ is isotropic and is a non-increasing function on $\text{dist}(\mathbf{x}_1, \mathbf{x}_2)$. Given $d, d' \in \mathbb{N}^*$, $\mathbf{X} := [\mathbf{x}_1, \dots, \mathbf{x}_d] \in \mathcal{D}^d$ and $\mathbf{X}' := [\mathbf{x}'_1, \dots, \mathbf{x}'_{d'}] \in \mathcal{D}^{d'}$, we write $f(\mathbf{X})$ and $k(\mathbf{X}, \mathbf{X}')$ as the $\mathbb{R}^d$ vector and $\mathbb{R}^{d \times d'}$ matrix respectively obtained by applying the function to each location point. Spatial Gaussian Process models are often equipped with a nugget to account for measurement error. We define the model as follows.

Definition 1 (Spatial Gaussian Process model with nugget). We assume that the observed value on each location $\mathbf{x} \in \mathcal{D}$ is the addition of the true value $z(\mathbf{x})$ with a Gaussian white noise ϵ , known as nugget.

$$\forall \mathbf{x} \in \mathcal{D}, u(\mathbf{x}) = z(\mathbf{x}) + \epsilon(\mathbf{x}) \quad (1)$$

where $z \sim \mathcal{GP}_{\mathcal{D}}(f, k)$ and $\epsilon(\mathbf{x}) \sim N(0, \sigma_{\epsilon}^2)$ is independent of $\mathbf{x}$. To specify the model without nugget, simply let $\sigma_{\epsilon}^2 = 0$.

Given a sample of location $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$, the model gives that

$$\mathbf{U} \sim \mathcal{N}_n(\boldsymbol{\mu}, \Sigma + \sigma_{\epsilon}^2 I_n) \quad (2)$$

where $\mathbf{U} = [u_1, \dots, u_n]^{\top}$ are the observations, I_n denotes the $n \times n$ identity matrix, $\boldsymbol{\mu} := f(\mathbf{X})$ and $\Sigma := k(\mathbf{X}, \mathbf{X})$. The predictive distribution of variables at new locations $\mathbf{X}^* = [\mathbf{x}_1^*, \dots, \mathbf{x}_m]$ is available as a closed-form multivariate Gaussian distribution (Williams and Rasmussen, 1995).

$$\mathbf{Z}^* | \mathbf{X}, \mathbf{U}, \boldsymbol{\theta} \sim \mathcal{N}_m(\boldsymbol{\mu}^*, \Sigma^*) \quad (3)$$

where

$$\begin{aligned} \boldsymbol{\mu}^* &:= f(\mathbf{X}^*) + k(\mathbf{X}^*, \mathbf{X})(\Sigma + \sigma_{\epsilon}^2)^{-1}(\mathbf{U} - \boldsymbol{\mu}) \\ \Sigma^* &:= k(\mathbf{X}^*, \mathbf{X}^*) - k(\mathbf{X}^*, \mathbf{X})(\Sigma + \sigma_{\epsilon}^2)^{-1}k(\mathbf{X}, \mathbf{X}^*) \end{aligned}$$

It involves a matrix inversion that requires $\mathcal{O}(n^3)$ time complexity, which is the computational bottleneck of GP models.

The mean function f and the covariance function k may be parameterised, denoted as $\{f_{\boldsymbol{\theta}_f} : \boldsymbol{\theta}_f \in \Theta_f\}$ and $\{k_{\boldsymbol{\theta}_k} : \boldsymbol{\theta}_k \in \Theta_k\}$. The full set of parameters are $\boldsymbol{\theta}_f, \boldsymbol{\theta}_k, \sigma_{\epsilon}^2$ in the space $\Theta := \Theta_f \times \Theta_k \times [0, \infty)$. The parameterised model is general enough to include the tasks of spatial interpolation and spatial regression. Inference on the parameters can be done in the frequentist approach with maximum likelihood estimation or in the Bayesian approach with a prior.

The covariance function k is often associated with two nonnegative parameters, the variance coefficient σ_k^2 that controls the overall magnitude of covariance, and the inverse range (aka. lengthscale) l that controls how covariance decreases along with distance. Two common choices of k are the RBF kernel and the Matérn kernel.

However, even with a large parameter space, it is common in real cases that data from different regions do not share the same parameter. We proceed to discuss some existing partition-based GP models that fit an individual GP in each local subspace.

3.2 Partition GP

The partition schemes on the space $\mathcal{D}$ have long been studied. It only requires that the assumptions (A1) and (A2) hold on smaller local subspaces. The prediction will be more efficient since the n in (2) is the sample size in the subspace. We first introduce some notation for partition. Let $P \in \mathbb{N}^*$ be the number of subspaces and $\mathcal{D}_p$ be the p -th subspaces such that $\bigsqcup_{p=1}^P \mathcal{D}_p = \mathcal{D}$. A partition is characterized as a membership function $m : \mathcal{D} \mapsto \mathbb{N}^*$, such that $m(\mathbf{x}) = p$ if and only if $\mathbf{x} \in \mathcal{D}_p$. The pair (P, m) uniquely identifies a partition.

Early work on partition-based GP focused on deterministic approaches, which divide $\mathcal{D}$ purely on the basis of observed location $\{\mathbf{x}_i\}_{i=1}^n$ before fitting the Gaussian model. For example, a simple approach is to divide the space with a grid with tunable size (Park et al., 2011). Shen et al. (2005) used an adaptive k-d tree to split the space recursively along one chosen data dimension each time. Park and Apley (2018) improves the flexibility by considering the Principal Component Vector at the midpoint as the split boundary.

The major drawback of deterministic approaches is that the fitness of the Gaussian model for y on each subspace is not considered. Probabilistic approaches remedy this issue by involving the partition scheme in the model. To the best of our knowledge, these models are built within the Bayesian framework. A prior is designed for (P, m) . Conditional on (P, m) , a GP model is built for each subspace with its own parameters. While the likelihood of a partition (P, m) is quantified, the normalizing factor of the posterior distribution is intractable. These approaches rely on MCMC to perform posterior sampling.

There are many schemes to specify the prior $p(P, m)$. Kim et al. (2005) proposed the Voronoi tessellation-based method. It models a set of subspace centres that are uniformly chosen from sample locations, such that a subspace is the set of points closer to its centre than other centres. Gramacy and Lee (2008) proposed the tree-GP model. It performs partitioning with a Bayesian probabilistic

classification and regression tree (CART), which is a binary tree whose nodes represent its subspace. The subspaces at the leaf nodes constitute the partition. The splitting criteria restrict that the subspaces are disjoint rectangles. Teixeira et al. (2019), Luo et al. (2021) and Luo et al. (2024) enhanced flexibility for the split by using spanning trees. Notably, Luo et al. (2024) constructed a weighted random graph upon a set of reference vertices from the sample location and partitioned the space by probabilistically cutting the edges of the spanning tree.

The posterior MCMC sampling for the above methods involves calculating an acceptance rate for each jump, which involves both the proposal distribution, the prior and the likelihood. It requires a costly computation of the inverse of the kernel matrix. The total time complexity is $\mathcal{O}(Bn^3)$, where B is the MCMC step size and n is the sample size. The tree-GP model in (Gramacy and Lee, 2008) took 60 hours to perform $B = 15,000$ MCMC steps on an $n = 3041$ sample. The high time complexity is the major drawback that makes these models infeasible on large samples.

3.3 Scalable GP with approximation

Our proposed methodology requires efficient GP fitting and evaluation, which can be realised with existing approximation approaches that induce sparsity to the matrix that requires inversion. (eg. Sparse Variational GP (Titsias, 2009), Vecchia approximation (Vecchia, 1988), Nearest neighbor GP (Datta et al., 2016), Hilbert space methods (Solin and Särkkä, 2020) etc.). In particular, Integrated Nested Laplace Approximation (INLA) (Lindgren et al., 2011) fits into our model settings.

INLA approximates GP with a Markov random field in the case that the covariance function k is a Matérn kernel with covariance coefficient σ_k^2 , lengthscale l and some specific choices of the smoothness parameter ν . For example, when $\nu = 1$, the following model is an approximation to $z \in \mathcal{GP}_{\mathcal{D}}(f, k)$ (Lindgren et al., 2011).

$$\begin{aligned} \forall \mathbf{x} \in \mathcal{D}, z(\mathbf{x}) &= f(\mathbf{x}) + \sum_{i=1}^n \phi_i(\mathbf{x}) w_i \\ \mathbf{w} &:= [w_1, \dots, w_n]^\top \sim \mathcal{N}_n(\mathbf{0}, \sigma_k^2 \mathbf{Q}(l)^{-1}) \end{aligned}$$

where

$$\mathbf{Q}(l) := \mathbf{K}(l)\mathbf{C}^{-1}\mathbf{K}(l)$$

$$\mathbf{K}(l) := l\mathbf{C} + \mathbf{G}$$

$$\mathbf{C}_{ij} := \langle \phi_i, \phi_j \rangle$$

$$\mathbf{G}_{ij} := \langle \nabla \phi_i, \nabla \phi_j \rangle$$

and $\phi_i : \mathcal{D} \mapsto [0, 1]$ are some basis functions dependent on the triangulation of $\mathcal{D}$ among the sample locations $\mathbf{X}$, such that $\phi_i(\mathbf{x}_j) = \mathbf{1}\{i = j\}$ and, therefore, $z(\mathbf{X}) = f(\mathbf{X}) + w(\mathbf{X})$. $\phi_i, \mathbf{C}, \mathbf{G}$ are fixed given the sample. $\mathbf{C}, \mathbf{G}, \mathbf{K}(l)$ are sparse matrix given its definition. $\mathbf{C}$ can be further approximated with a diagonal matrix. Thus, the precision matrix $\mathbf{Q}$ is also sparse, and the numerical factorization of $\mathbf{Q}$ is available in $\mathcal{O}(n^{3/2})$ time (Lindgren et al., 2011). With further Laplace approximation, posterior and predictive inference can be performed by sampling in $\mathcal{O}(n^{3/2})$. (Bakka et al., 2018). Additionally, we can use the Woodbury matrix identity (Woodbury, 1950) to derive the gradients of the log likelihood of the observation $u(\mathbf{X}) \sim \mathcal{N}_n(f(\mathbf{X}), \sigma_k^2 \mathbf{Q}(l)^{-1} + \sigma_\epsilon^2)$ with respect to the parameters, which are available in $\mathcal{O}(n^2)$. The MLE with gradient methods with G iterations is achievable in $\mathcal{O}(Gn^2)$.

4 Methodology

To construct a computationally efficient model, we propose to avoid the expensive MCMC steps and design algorithms that partition the space $\mathcal{D}$ adaptively according to some metrics. The algorithms will be iterative. Given a current partition (P, m) on the sample $(\mathbf{X}, \mathbf{U})$, we compute a metric $L(P, m, \mathbf{X}, \mathbf{U})$ that measures the fitness of the model. The choices of L will be discussed in Section 4.3. We decide whether there is a need to further propose a new partition. If this is true, a new partition (P', m') will be proposed by a greedy partition scheme. Otherwise, the iteration terminates. $\mathcal{A}(L, \mathbf{X}, \mathbf{U}, t)$ represents such decision function. Algorithm 1 describes the framework of training a partition-based GP model. t is the index of the current iteration. We may impose a budget T for the maximum iteration steps.

The following discusses the details about the partition scheme for line 5, the approximation based scalable GP fitting for line 6, and the choices of metrics L for line 7. Under this framework, We

Algorithm 1 Partition-Based GP Training Framework

- 1: Set $t \leftarrow 0, P \leftarrow 1, m(\mathbf{x}) \leftarrow 1 \forall \mathbf{x} \in \mathcal{D}$.
 - 2: Fit a GP on $\mathcal{D}$.
 - 3: Compute $L(P, m, \mathbf{X}, \mathbf{U})$
 - 4: **while** $\mathcal{A}(L, \mathbf{X}, \mathbf{U}, t)$ is TRUE **do**
 - 5: Propose a new partition (P', m') based on (P, m) and $L(P, m, \mathbf{X}, \mathbf{U})$
 - 6: Fit an individual GP on each subspace in (P', m') .
 - 7: Compute $L(P', m', \mathbf{X}, \mathbf{U})$
 - 8: Update $(P, m) \leftarrow (P', m'), t \leftarrow t + 1$
 - Terminate**: report (P, m) and GPs on each subspace.
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propose the INLA-tree-GP model that partitions the space with a decision tree and uses the INLA approach to fit a GP.

4.1 Partition scheme

Inspired by the Bayesian tree-GP model (Gramacy and Lee, 2008) described in section 3.2, we propose a partition scheme based on a decision tree. As in the tree-GP model, each node η in the tree $\mathcal{T}$ still represents a rectangular subspace, and the leaf nodes represent the partition. The tree is built by the standard decision tree learning process, such that a split on a leaf node is performed in each iteration only if it has the maximum gain in model fitness, measured by

$$L(\mathcal{T}', \mathbf{X}, \mathbf{U}) - L(\mathcal{T}, \mathbf{X}, \mathbf{U}) := L(\eta_L, \mathbf{X}, \mathbf{U}) + L(\eta_R, \mathbf{X}, \mathbf{U}) - L(\eta, \mathbf{X}, \mathbf{U}) \quad (4)$$

where η is a leaf node, η_L, η_R are the proposed new children, and $\mathcal{T}' := \mathcal{T} \cup \{\eta_L, \eta_R\}$ is the new tree. The split is restrictive to the lines parallel to the axis of dimension $d \in \{1, 2\}$ at a sample location $\mathbf{x}_d \in \mathbf{X}$. The new children represent the subspaces

$$\mathcal{D}^{\eta_L} := \{\mathbf{x} \in \mathcal{D}^\eta : \mathbf{x}_d \leq x\}, \quad \mathcal{D}^{\eta_R} := \{\mathbf{x} \in \mathcal{D}^\eta : \mathbf{x}_d > x\}$$

To find the best split, we enumerate all positions in $\mathbf{X}$ and both coordinates $d \in \{1, 2\}$. This accounts for a time complexity of $\mathcal{O}(n)$ for each iteration. The total time complexity will be $\mathcal{O}(Tn \times (\text{time for GP fit} + \text{time for computing } L))$. Since the iteration T is also the number of subspaces, we assume it to be small, as we require each subspace containing enough sample locations for model

fitting and too many subspaces suggest overfitting. In the practice of using the tree-GP, T is generally less than 20 (Gramacy and Lee, 2008; Luo et al., 2024).

Note that the split is still restricted to the lines parallel to each coordinate. The scheme suffers from inflexibility, the same issue as tree-GP in Gramacy and Lee (2008). Potentially, we will explore more partition schemes that allow for more flexible splits. For example, we may loosen the restriction to any line in $\mathbb{R}^2$. We might potentially design some greedy approach based on auxiliary graphs and spanning trees as in (Luo et al., 2024).

4.2 Efficient GP fitting

We need an efficient approach to fit a GP model for each subspace. We propose fitting the GP by obtaining the MLE or posterior statistics (eg, posterior mean, posterior median, Maximum a posteriori estimation) of the parameters. A scalable GP approximation approach is required to circumvent the costly matrix inversion and obtain the gradient efficiently.

The requirements on the approximation approach depend on the partition scheme. In our decision tree-based model, we propose to use the INLA approach described in Section 3.3, since it possesses the following properties.

1. Posterior statistics can be estimated by sampling in $\mathcal{O}(mn^{3/2})$, where m is the posterior sample size.
2. MLE of the parameters are available in $\mathcal{O}(Gn^2)$, where G is the iteration of the gradient method.
3. ϕ_i , $\mathbf{C}$, $\mathbf{G}$ are fixed given the samples, and restricting them to a subspace amounts to taking the submatrix. It allows for pre-processing and amortising the cost.
4. We expect an extra benefit in computation for the model without nugget because the formulas of the gradients are further simplified. Furthermore, since $\mathcal{D}^{n_L}$ and $\mathcal{D}^{n_R}$ will be monotonic as we enumerate the splitting positions, online methods for fitting without nugget may be available to further enhance scalability.

4.3 Metrics for fitness

The fitness metric L of the overall model can be designed as the sum (or some other aggregation) of the individual GP fitness, that is,

$$L(P, m, \mathbf{X}, \mathbf{U}) = \sum_{p=1}^P L(\mathcal{GP}_{\mathcal{D}_p}, \mathbf{X}^p, \mathbf{U}^p) \quad (5)$$

where $\mathcal{GP}_{\mathcal{D}_p}$ is the fitted Gaussian Process in the p -th subspace and $(\mathbf{X}^p, \mathbf{U}^p)$ is the sample in $\mathcal{D}_p$. In the INLA-Tree-GP model, it suffices to compare the L before a split and after a split as in formula (4). We list some choices of L as follows.

1. Mean squared error on a holdout set: This is the most straightforward and computationally efficient way. However, the evaluation greatly depends on the locations of the holdout set, and it requires a relatively large sample in each subspace.
2. Likelihood: It directly measures the strength of fitness without losing sample data. However, a fast computation of likelihood or log likelihood is required. The INLA fails regarding this. Additionally, it does not account for the variability outside the sample.
3. Integration of predictive variance: It is estimated by summing up the predictive variance on a prespecified mesh. The model will target to reduce the uncertainty throughout the space, rather than to enhance the fitness on the sample. But it may be computationally expensive if the mesh is fine.
4. Aggregation of 1 and 3 or 2 and 3: It accounts for both fitness on the sample and uncertainty outside the sample. Computation efficiency may still be an issue.

We evaluate the time complexity of the INLA-tree-GP model to be $\mathcal{O}(Tmn^{5/2})$ (using INLA posterior sampling) or $\mathcal{O}(Tn^3)$ (using MLE with gradient approach). The value of T can be as small as 20 even for a large sample. While the scale of m and G remain to be tested, we are optimistic that Tm and TG can be much smaller than the MCMC steps B .

5 Anticipated outcomes

We will develop partition-based spatial Gaussian Process models within a frequentist framework as describe in Algorithm 1. The INLA-tree-GP model will be a promising example that is more computationally efficient to the traditional Bayesian approaches. The model will be implemented in an R or Python package. We anticipate a better performance in fitness and scalability than the current benchmarks. Future work includes developing models that is less restrictive on the partitioning boundaries.

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